

# Lecture II – Explore Vs. Exploit

## Programming: Everyday Decision-Making Algorithms

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### Explore Vs. Exploit

You have 45 minutes for lunch. Do you try the new restaurant or go to your reliable favorite?

...

🗳️ Quick Poll: Raise your hand, who would try the new restaurant?

...

This decision happens hundreds of times daily — and there's optimal math behind it!

Photo by Jillian Amatt on Unsplash

### Core Concepts

#### 🔍 Explore

- **Definition:** Gathering new information
- **Risk:** Unknown outcomes
- **Example:** Trying that new sushi place

#### 🎯 Exploit

- **Definition:** Using known information
- **Risk:** Missing better options
- **Example:** Your go-to pizza joint

#### ⚖️ The Fundamental Trade-off

These are **mutually exclusive** choices in any given moment. You can't have your cake and eat it (too).

**Question: Do you have any examples of explore vs exploit in your life?**

#### 🏥 Medical: Clinical Trials

**Explore:** Test experimental drug on new patients

- *Risk*: Unknown side effects, potential harm
- *Reward*: Breakthrough treatment discovery

**Exploit:** Use proven treatment

- *Risk*: Missing better cures
- *Reward*: Predictable, safe outcomes

## 🖥️ Tech: A/B Testing

**Explore:** Test new website design

- *Cost*: Development time, potential lost conversions
- *Benefit*: Higher conversion rates, better UX

**Exploit:** Keep current design

- *Cost*: Missed optimization opportunities
- *Benefit*: Stable, known performance

## 💕 Personal: Dating Apps

**Explore:** Meet someone new

- *Risk*: Bad dates, wasted time
- *Reward*: Finding "the one"

**Exploit:** Meet someone you already know

- *Risk*: Settling for "good enough"
- *Reward*: Comfortable, known connection

## The Daily Decision Dilemma

**Explore Moments**

- Try a new coffee shop
- Learn a new skill
- Watch a different genre
- Take a new route to work

**Exploit Moments**

- Order your usual drink
- Use familiar tools
- Rewatch favorite shows
- Take the fastest route

💡 The Million Dollar Question

**What is the optimal balance?** Science has been asking this for **over 50 years!**

## The problem with exploitation

**Question: Any ideas what it might be?**

- Exploitation is not always the best strategy.
- Especially when you have **very limited information**.
- When you **stop exploring**, you might **miss better options**.
- Imagine you are not able to gather new information and could **only choose known alternatives**.

## The problem with exploration

**Question: Any ideas what it might be?**

- Exploration is not always the best strategy.
- Especially when you are **limited in using new information**.
- When you **constantly explore**, you might **never enjoy the fruits of your exploration**.
- Imagine you could only **eat each meal only once, hear each song only once, talk to each person only once...**

## Multi-Armed Bandit Problem

### Decision Support

- To provide support, computer scientists formulated the explore-exploit trade-off.
- It is known as the multi-armed bandit problem.
- Question: What is a one-armed bandit?

### One Armed Bandit

Photo by Kabir M on Unsplash

### Multi-Armed Bandit Problem

**Imagine:** You walk into a casino with **3 slot machines**

- Each machine has a **different** (unknown) win rate
- You have **\$100** and **20 pulls** total
- **Goal:** Maximize your winnings

**How is this related to explore vs. exploit?**

- **Explore:** Try different machines
- **Exploit:** Play the best-performing machine

### Multi-Armed Bandit Problem II

- By playing a slot machine, the gambler can gather information about the probability of winning.
- But each pull of a lever comes with a certain cost.
- It's the aim of the gambler to maximize his winnings.

## The Core Question

How do you decide which machine to play next?

## Interactive Casino Challenge

### Your Current Situation

You've been playing for a while. Here's what you know:

| Machine    | Pulls | Wins | Win Rate | Your Choice? |
|------------|-------|------|----------|--------------|
| <b>A</b> 🎰 | 15    | 9    | 60%      |              |
| <b>B</b> 🎰 | 2     | 1    | 50%      |              |
| <b>C</b> 🎰 | 0     | 0    | ???      |              |

💡 Class Vote: Raise your hand for your choice!

- **Machine A** (known performer)
- **Machine B** (limited data)
- **Machine C** (complete unknown)

## Casino Challenge Follow-up

### Your Current Situation

You've been playing for a while. Here's what you know:

| Machine    | Pulls | Wins | Win Rate | Your Choice? |
|------------|-------|------|----------|--------------|
| <b>A</b> 🎰 | 15    | 9    | 60%      |              |
| <b>B</b> 🎰 | 2     | 1    | 50%      |              |
| <b>C</b> 🎰 | 0     | 0    | ???      |              |

i Follow-up Questions

- What if you only had **2 pulls left**?
- What if you had **100 pulls left**?
- How does this change your strategy?

## Expected value for a decision

- The expected value of a slot machine is the average number of wins per pull.
- Expected value of machine A =  $E(A) = 9/15 = 0.6$
- Expected value of machine B =  $E(B) = 1/2 = 0.5$
- Machine A has the higher expected value.
- But 2 and even 15 pulls are not a large number (considering standard deviation).

### Multi-Armed Bandit III

- The multi-armed bandit problem represents a lot of different real-world decisions.
- It shows us, that there might be a difference between the optimal long-term average performance and the optimal immediate performance.
- Which lever to pull next depends completely on something we haven't discussed yet:

### Multi-Armed Bandit IV

- How long you plan to be in the casino?
- Question: Why is this important?
- Question: How does this influence our decision on taking machine A or machine B?

...

"I'm more likely to try a new restaurant when I move to a city than when I'm leaving it" (Chris Stucchio)

### The influence of the interval

- Let's call the time you plan to be in the casino "the interval".
- The longer the interval, the more (in general) you should explore, since you will have more opportunities to exploit the gathered information.
- The shorter the interval, the more you should exploit your current knowledge.
- The optimal strategy depends on the length of the interval.

### Interval and Exploration

Explore when you have the time to use the resulting knowledge.

"I moved to Pune, India, and I just [...] eat everywhere that didn't look like it was gonna kill me" (Chris Stucchio)

### Exploration

Photo by Colin Maynard on Unsplash

### Interval and Exploitation

Exploit when you are ready to cash in.

"And as I was leaving the city I went back to all my old favorites, rather than trying out new stuff [...]. Even if I find a slightly better place, I'm only going to go there once or twice, so why take the risk?" (Chris Stucchio)

### Exploitation

Photo by Cristina Gottardi on Unsplash

## Hollywood's Strategy

### Sequels in Top 10 Highest-Grossing Movies

| Year | Number of Sequels |
|------|-------------------|
| 1981 | 2                 |
| 1991 | 3                 |
| 2001 | 5                 |
| 2011 | 8                 |

### Class Discussion

#### What do you think:

- What explains this trend?
- How does this relate to explore vs. exploit?
- What does this suggest about Hollywood's "time horizon"?

## Sequels...

...

The Insight: Studios are **exploiting** proven franchises because they think they're approaching the **"end of their interval"** (streaming disruption, changing audience preferences).

Photo by Universal Pictures on Wikipedia

## A Possible Explanation

- Making a brand new movie is risky but has the potential to create a new fan base. (explore)
- From a Studio's perspective, a sequel is a movie with a guaranteed fan base, a cash cow, a sure thing, an exploit.
- One possible explanation for the numbers is that the studios think they are approaching the end of their interval.
- They are pulling the arms of the best machines they've got before the casino turns them out.

## Multi-Armed Bandit VI

- While the so far provided anecdotes are helping us to understand the explore-exploit trade-off, they are far away from being a satisfying "optimal" solution.
- Actually, finding an algorithm that tells us exactly how to handle the trade-off is a very hard problem.
- On the way there were many interesting approaches...

## Win-Stay Lose-Shift

### Win-Stay Lose-Shift<sup>1</sup>

- Question: What do you think, what the win-stay lose-shift strategy does?
- The win-stay lose-shift strategy is a simple strategy that is often used in multi-armed bandit problems.
- Based on the idea that if you have won, you should stay. If you have lost, you should switch.
- This strategy is not always the best strategy, but it is simple and proven better than choose an arm at random.

### Win-Stay = No Brainer?

- Question: What do you think about win-stay?
- If you decide to pull an arm and you win, you should pull the same arm again.
- Nothing changes, except the attractiveness of the arm you just pulled is higher.

### Lose-Shift is another story

- Question: What do you think about lose-shift?
- Changing arms each time you lose might be a pretty rash move.
- Imagine you're eating at your favorite restaurant for the tenth time in a row.
- You have always been very satisfied (win), but today you are disappointed (lose).
- Should you turn your back on the restaurant?

### Like most of the time, easy answers comes with problems

- The win-stay lose-shift strategy penalizes losses too much.
- The strategy does not take into account the interval.
- But the strategy was good start to develop better approaches.

## The Bellman Approach

### The Bellman approach I

- Few years later, Richard Bellman, found an exact solution for all cases with finite and known intervals.
- Bellman found that under assumptions, exploit vs explore can be seen as an optimal stopping problem.
- The question is, when to stop exploring and start exploiting.
- The solution is based on dynamic programming and backward calculation starting from the final pull (analogous to the secretary problem with full information).

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<sup>1</sup>For more details see Robbins, H. (1952) 'Some aspects of the sequential design of experiments', Bulletin of the American Mathematical Society, 58.

## The Bellman approach II

Bellman found that the **following equation** provides the optimal strategy (when the assumptions hold):

$$\mathbb{E}[\cdot] = \frac{w + 1}{w + l + 2}$$

where  $w$  is the number of wins and  $l$  is the number of losses.

## Question time

$$\mathbb{E}[\cdot] = \frac{w + 1}{w + l + 2}$$

- Question: What is  $E[B]$ ?
- Question: What is  $E[A]$ ?
- Question: What machine shall we play according to the Bellman approach?

## The Bellman approach III

The following table shows the expected value for different win-lose scenarios.

| Loses/Wins | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
|------------|------|------|------|------|------|------|------|------|------|------|
| 1          | 0.50 | 0.60 | 0.67 | 0.71 | 0.75 | 0.78 | 0.80 | 0.82 | 0.83 | 0.85 |
| 2          | 0.40 | 0.50 | 0.57 | 0.63 | 0.67 | 0.70 | 0.73 | 0.75 | 0.77 | 0.79 |
| 3          | 0.33 | 0.43 | 0.50 | 0.56 | 0.60 | 0.64 | 0.67 | 0.69 | 0.71 | 0.73 |
| 4          | 0.29 | 0.38 | 0.44 | 0.50 | 0.55 | 0.58 | 0.62 | 0.64 | 0.67 | 0.69 |
| 5          | 0.25 | 0.33 | 0.40 | 0.45 | 0.50 | 0.54 | 0.57 | 0.60 | 0.63 | 0.65 |
| 6          | 0.22 | 0.30 | 0.36 | 0.42 | 0.46 | 0.50 | 0.53 | 0.56 | 0.59 | 0.61 |
| 7          | 0.20 | 0.27 | 0.33 | 0.38 | 0.43 | 0.47 | 0.50 | 0.53 | 0.56 | 0.58 |
| 8          | 0.18 | 0.25 | 0.31 | 0.36 | 0.40 | 0.44 | 0.47 | 0.50 | 0.53 | 0.55 |
| 9          | 0.17 | 0.23 | 0.29 | 0.33 | 0.38 | 0.41 | 0.44 | 0.47 | 0.50 | 0.52 |
| 10         | 0.15 | 0.21 | 0.27 | 0.31 | 0.35 | 0.39 | 0.42 | 0.45 | 0.48 | 0.50 |

## The Bellman approach IV

- The calculation of the optimal strategy is very extensive when there are many arms and a long interval.
- And yet the approach does not help us in many scenarios because we do not know the exact length of the interval (time in the casino).
- At this point, it looked like the multi-armed bandit problem would remain a problem without a solution.

## The Gittins Index

### The Gittins Index I<sup>2</sup>

- In the 1970s Unilever asked a young mathematician, John Gittins, to help optimize their drug trials.
- Given different compounds, what is the quickest way to determine which is likely to be effective?
- Gittins found an optimal strategy and abstracted the problem to a general level.
- He found the solution to the multi-armed bandit problem.

### The Gittins Index II

- A major problem with the multi-armed banded problem is that previous solutions made very critical assumptions about the underlying interval.
- For example, that the length of the interval is known at the beginning of the analysis.
- In his approach, future wins (e.g., cash flows) are discounted so that any interval length (including infinity) can be considered.
- Geometric discounting assumption, but the approach can be extended to other discounting assumptions.

### What is Discounting?

- Future rewards are valued less than immediate rewards.
- Future rewards are weighted by a discount factor.
- **Monetary context:** Easy to justify discounting:
  - Interest rates: Money today can earn interest.
  - Opportunity costs: Immediate use of money has value.
- **Non-monetary context:** Discounting also makes sense:
  - Time: Immediate gratification often preferred.
  - Example: Tonight's dinner vs. the same dinner in a week's time.

### The Gittins Index III

- The Gittins index can be used for any problems of the form of the multi-armed bandit problem.
- That means it solves the explore-exploit trade-off.
- Let's consider our machine A and B example one last time.
  - Machine A: 15 pulls, 9 wins, 6 loses.
  - Machine B: 2 pulls, 1 win, 1 lose.

### The Gittins Index IV

| Loses/Wins | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0          | .7029 | .8001 | .8452 | .8723 | .8905 | .9039 | .9141 | .9221 | .9287 | .9342 |

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<sup>2</sup>For more details see Gittins, J. (1979) 'Bandit Processes and Dynamic Allocation Indices', Journal of the Royal Statistical Society. Series B (Methodological).

| Loses/Wins | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1          | .5001 | .6346 | .7072 | .7539 | .7869 | .8115 | .8307 | .8461 | .8588 | .8695 |
| 2          | .3796 | .5163 | .6010 | .6579 | .6996 | .7318 | .7573 | .7782 | .7956 | .8103 |
| 3          | .3021 | .4342 | .5184 | .5809 | .6276 | .6642 | .6940 | .7187 | .7396 | .7573 |
| 4          | .2488 | .3720 | .4561 | .5179 | .5676 | .6071 | .6395 | .6666 | .6899 | .7101 |
| 5          | .2103 | .3245 | .4058 | .4677 | .5168 | .5581 | .5923 | .6212 | .6461 | .6677 |
| 6          | .1815 | .2871 | .3647 | .4257 | .4748 | .5156 | .5510 | .5811 | .6071 | .6300 |
| 7          | .1591 | .2569 | .3308 | .3900 | .4387 | .4795 | .5144 | .5454 | .5723 | .5960 |
| 8          | .1413 | .2323 | .3025 | .3595 | .4073 | .4479 | .4828 | .5134 | .5409 | .5652 |
| 9          | .1269 | .2116 | .2784 | .3332 | .3799 | .4200 | .4548 | .4853 | .5125 | .5373 |

## Question

**Choose machine A or B according to the Gittins index?**

- The index for machine B (0.6346) is higher than for machine A (0.6300).
- The index shows a clear win-stay pattern.
- There is a relaxed lose-shift pattern.
- At the (0,0) point we see the exploration bonus (premium).
- The index converges to 1/2 for a 50/50 chance game.

## Gittins Index: The Scary Math

### 🧐 The Complex Formula

$$G_i(s_i, f_i) := \sup_{\tau \geq 1} \frac{\mathbb{E} \left[ \sum_{t=0}^{\tau-1} \beta^t \cdot r_i^t \mid s_i, f_i \right]}{\mathbb{E} \left[ \sum_{t=0}^{\tau-1} \beta^t \right]}$$

- **Extremely complex** to calculate by hand
- **Requires advanced mathematics** (optimization theory)
- **Calculation not practical** for everyday decisions

! Important

**You don't need to understand the math** - just use the lookup table!

## What It Actually Means

- **Core idea:** "What's the best possible average reward from this option?"
- **Higher Index = Better Choice**
- **Example:** Machine A (9 wins, 6 losses): Index = 0.6300; Machine B (1 win, 1 loss): Index = 0.6346
- **Machine B wins** due to exploration bonus for less-tested options
- The Gittins Index automatically balances exploration and exploitation

## Explore vs Exploit: Summary

### Explore vs Exploit: Summary

- Consider an explore vs exploit decision situation.
- As you learned exploiting comes with a known (expected) outcome for example  $E(\text{exploit}) = X$
- Exploring comes with an unknown outcome  $E(\text{explore}) = ?$
- What should you do according to decision science?

### Explore vs Exploit: Anecdotally

- If you have a long interval, you should explore, choose B until you are sure about  $E(\text{explore})$ .
- If you have a short interval, you should exploit, choose A.

### Explore vs Exploit: Mathematically

- If  $E(\text{exploit})$  and  $E(\text{explore})$  are known, choose higher expected value.
- If  $E(\text{explore})$  is unknown, but you know the length of the interval, the Bellman-approach provides the optimal strategy.
- If  $E(\text{explore})$  is unknown, and you do not know the length of the interval, the Gittins index provides the optimal strategy.

## Your Personal Action Plan

 Rate yourself (1-10) on each:

- **Career exploration:** Trying new skills/roles vs. deepening current expertise
- **Social exploration:** Meeting new people vs. investing in current relationships
- **Learning exploration:** New subjects vs. mastering current interests
- **Life exploration:** New experiences vs. enjoying familiar pleasures

### Key Insights to Remember

1. **Exploration has inherent value** - it increases your chance of finding the best
2. **Time horizon matters** - more time left = more exploration optimal
3. **Age affects strategy** - younger people should explore more
4. **Don't stop exploring completely** - even experts need some exploration

## Takeaways

Your today's takeaway from the lecture should be: Be sensitive to how much time you have left in the casino and **explore, explore, explore...**

# Literature

## Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.<sup>3</sup>
- Ferguson, T.S. (1989) 'Who solved the secretary problem?', Statistical Science, 4(3). doi:10.1214/ss/1177012493.

## Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Here](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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### Note

Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

## More Literature

For more interesting literature, take a look at the [literature list](#) of this course.

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<sup>3</sup>The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.